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3.2 Light

Cambridge IGCSE Physics (0625) | Syllabus 2026–2028

Covers: 3.2.1 Reflection | 3.2.2 Refraction | 3.2.3 Thin lenses | 3.2.4 Dispersion

Tier: Core and Extended

Name: _____ Class: _____ Date: _____

How to use this booklet

1. Rate every syllabus statement Red, Amber or Green now, then again after you have worked through the booklet.
2. Draw the diagrams in Section 4 yourself. Drawing and labelling them is a skill the exam tests, so a blank box here is deliberate.
3. Answer the practice questions before you look at the mark scheme. Your teacher has the mark scheme and model diagrams.

Section 1: Syllabus checklist

Blue = Core | Amber = Extended (Supplement)

#	Syllabus statement (verbatim)	Tier	R	A	G
1	Define and use the terms normal, angle of incidence and angle of reflection	Core			
2	Describe the formation of an optical image by a plane mirror and give its characteristics, i.e. same size, same distance from mirror, virtual	Core			
3	State that for reflection, the angle of incidence is equal to the angle of reflection; recall and use this relationship	Core			
4	Use simple constructions, measurements and calculations for reflection by plane mirrors	Ext			
5	Define and use the terms normal, angle of incidence and angle of refraction	Core			
6	Describe an experiment to show refraction of light by transparent blocks of different shapes	Core			
7	Describe the passage of light through a transparent material (limited to the boundaries between two mediums only)	Core			
8	State the meaning of critical angle	Core			
9	Describe internal reflection and total internal reflection using both experimental and everyday examples	Core			
10	Define refractive index, n, as the ratio of the speeds of a wave in two different regions	Ext			
11	Recall and use the equation n = sin i / sin r	Ext			
12	Recall and use the equation n = 1 / sin c	Ext			
13	Describe the use of optical fibres, particularly in telecommunications	Ext			
14	Describe the action of thin converging and thin diverging lenses on a parallel beam of light	Core			

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#	Syllabus statement (verbatim)	Tier	R	A	G
15	Define and use the terms focal length, principal axis and principal focus (focal point)	Core			
16	Draw and use ray diagrams for the formation of a real image by a converging lens	Core			
17	Describe the characteristics of an image using the terms enlarged/same size/diminished, upright/inverted and real/virtual	Core			
18	Know that a virtual image is formed when diverging rays are extrapolated backwards and does not form a visible projection on a screen	Core			
19	Draw and use ray diagrams for the formation of a virtual image by a converging lens	Ext			
20	Describe the use of a single lens as a magnifying glass	Ext			
21	Describe the use of converging and diverging lenses to correct long-sightedness and short-sightedness	Ext			
22	Describe the dispersion of light as illustrated by the refraction of white light by a glass prism	Core			
23	Know the traditional seven colours of the visible spectrum in order of frequency and in order of wavelength	Core			
24	Recall that visible light of a single frequency is described as monochromatic	Ext			

Section 2: Key definitions

Term	Definition
Normal	A line drawn at 90° to a surface, at the point where a ray meets the surface
Angle of incidence (i)	The angle between the incident ray and the normal
Angle of reflection	The angle between the reflected ray and the normal
Angle of refraction (r)	The angle between the refracted ray and the normal
Critical angle (c)	The angle of incidence in the optically denser medium for which the angle of refraction is 90°
Total internal reflection	All of the light is reflected back into the denser medium. It happens when the angle of incidence in that medium is greater than the critical angle, so there is no refracted ray
Principal axis	The straight line through the optical centre of a lens, at 90° to the lens
Principal focus (focal point)	For a thin converging lens, the point on the principal axis where rays parallel to the principal axis meet after passing through the lens. For a thin diverging lens, the point on the incident side from which those rays appear to have come
Focal length (f)	The distance from the optical centre of a lens to the principal focus
Real image	An image formed where rays actually meet. It can be projected on a screen

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Term	Definition
Virtual image	An image formed by extrapolating diverging rays backwards. It does not form a visible projection on a screen
Dispersion	The separation of white light into the colours of the spectrum when the light is refracted by a glass prism. Violet is refracted more than red
Refractive index (n)	The ratio of the speeds of a wave in two regions. For light going from air into a medium, n = speed in air / speed in the medium. n has no unit
Monochromatic light	Visible light of a single frequency. A single wavelength says the same thing. Naming one colour is not enough, because one colour can cover a range of frequencies

Section 3: Formulae and equations

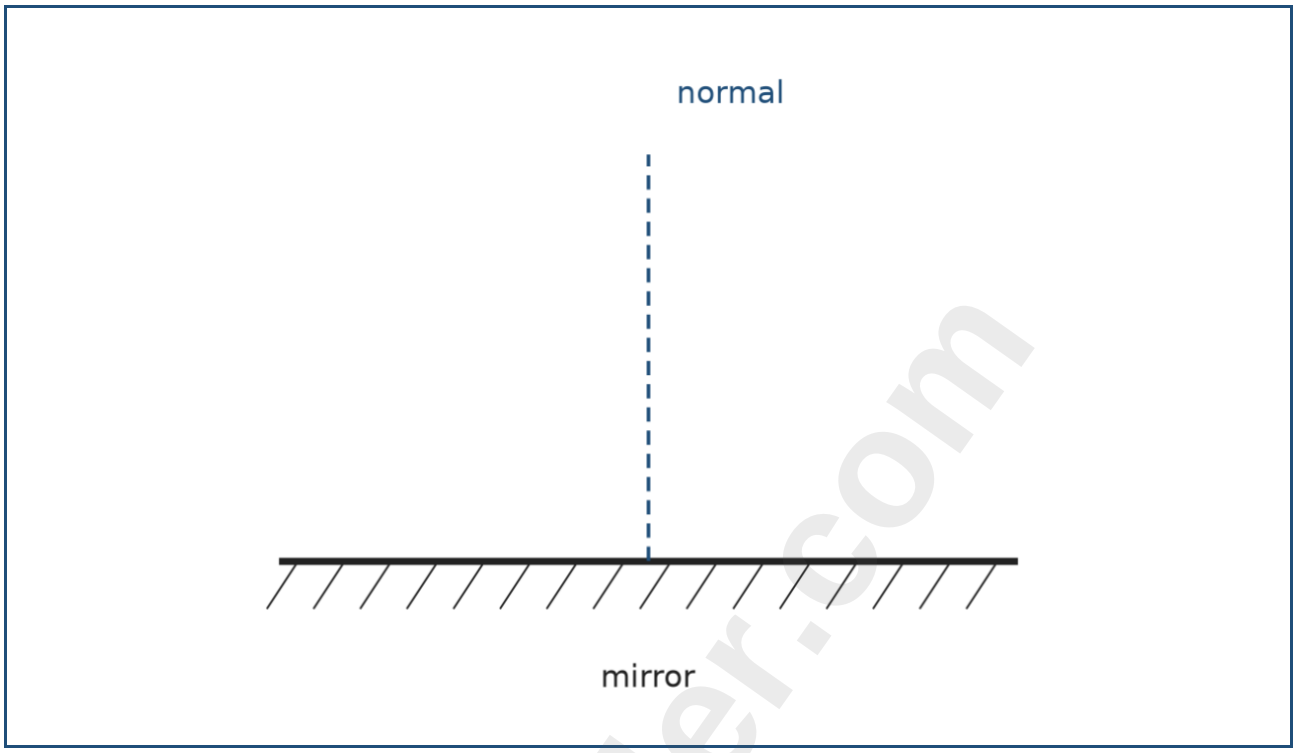
Quantity	Equation	Symbols and units
Law of reflection	**i = r**	i = angle of incidence, r = angle of reflection. Both are measured from the normal, not from the mirror
Refractive index (speeds)	**n = speed in air / speed in the medium**	n has no unit. Speed in the medium = speed in air / n. Use m/s for both speeds
Refractive index (angles)	**n = sin i / sin r**	i = angle of incidence in air, r = angle of refraction in the medium. Both from the normal. This form is for a boundary with air
Critical angle	**n = 1 / sin c**	c = critical angle in the medium, n = refractive index of that medium. The other side of the boundary is air. So sin c = 1 / n

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Section 4: Diagrams to draw

4.1 Reflection at a plane mirror Core

Task: Draw an incident ray that meets the mirror, and the reflected ray. Mark the angle of incidence i and the angle of reflection r, each measured from the normal. Make the two angles equal.

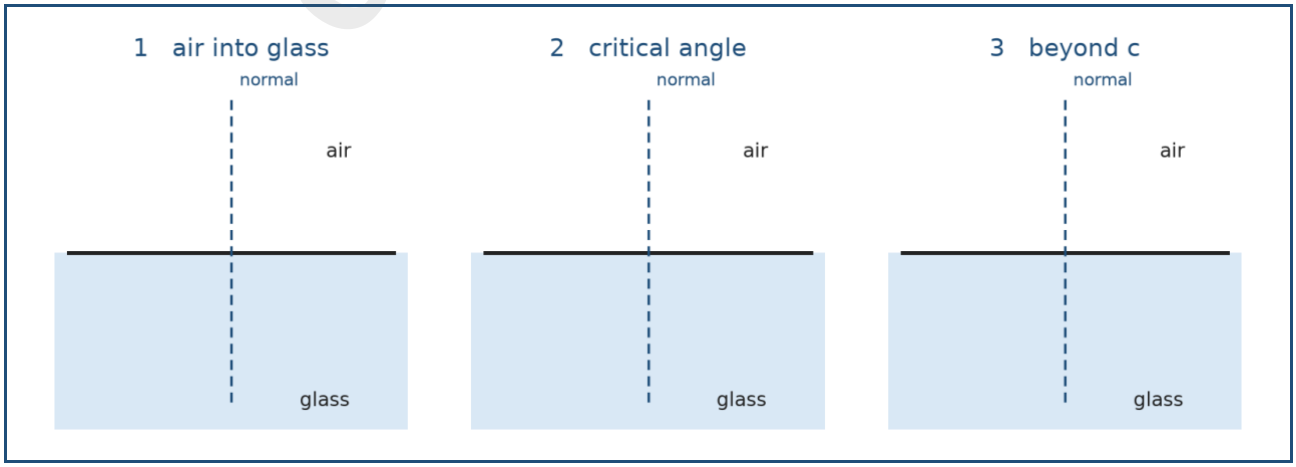


Check your diagram has:

- ☐ Arrow on the incident ray, towards the mirror
- ☐ Arrow on the reflected ray, away from the mirror
- ☐ i between the incident ray and the normal
- ☐ r between the reflected ray and the normal
- ☐ i equal to r

4.2 Refraction and total internal reflection Core

Task: Complete the three sketches. (1) A ray in air entering the glass: draw the refracted ray, with the angle of refraction smaller than the angle of incidence. (2) The critical-angle case: the refracted ray travels along the boundary. Label c. (3) An angle of incidence greater than the critical angle: draw the reflected ray inside the glass and do not draw a ray leaving the glass.



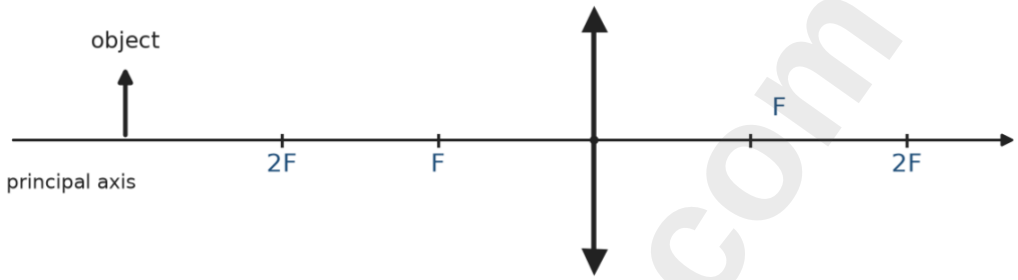
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Check your diagram has:

- ☐ Sketch 1: ray bends towards the normal as it enters the glass
- ☐ i and r both measured from the normal, with r smaller than i
- ☐ Sketch 2: refracted ray along the boundary and c labelled
- ☐ Sketch 3: reflected ray only, with the angle of reflection equal to the angle of incidence
- ☐ Arrows showing the direction of each ray

4.3 Real image formed by a converging lens Core

Task: The object is further from the lens than 2F. Draw two rays from the top of the object to locate the image, and draw the image arrow.

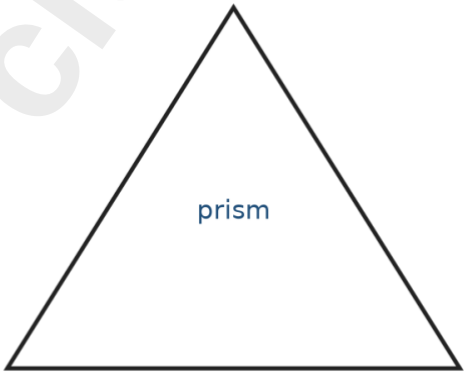


Check your diagram has:

- ☐ One ray parallel to the principal axis, refracted through F on the far side
- ☐ One ray through the optical centre, not bent
- ☐ Image arrow where the two rays meet
- ☐ Image inverted, between F and 2F on the opposite side to the object
- ☐ Image shorter than the object

4.4 Dispersion of white light by a prism Core

Task: Draw a narrow beam of white light entering one face of the prism, and the spectrum leaving the other face. Label the seven colours in order, from the colour refracted least to the colour refracted most.



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Check your diagram has:

- ☐ White light labelled on the way in
- ☐ The beam changes direction at each face
- ☐ Seven colours in order: red, orange, yellow, green, blue, indigo, violet
- ☐ Red labelled as refracted least (longest wavelength, lowest frequency)
- ☐ Violet labelled as refracted most (shortest wavelength, highest frequency)

Section 5: Worked examples

Worked Example 1 Extended

A ray of light in air meets a flat glass surface. The angle of incidence is 40° and the angle of refraction is 25°. Calculate the refractive index of the glass.

n = sin i / sin r

n = sin 40° / sin 25°

n = 0.6428 / 0.4226

n = 1.52

n = 1.52 (no unit)

Worked Example 2 Extended

The refractive index of a glass fibre is 1.50. Calculate the critical angle for light travelling in this glass and meeting air.

n = 1 / sin c

sin c = 1 / n = 1 / 1.50 = 0.6667

$c = \sin^{-1}(0.6667) = 41.8°$

c = 41.8°. Total internal reflection happens if the angle of incidence in the glass is greater than 41.8°.

Worked Example 3 Core

A pin stands 15 cm in front of a plane mirror. State three characteristics of the image, including where it is.

The image is the same size as the pin.

The image is the same distance behind the mirror as the pin is in front, so it is 15 cm behind the mirror.

The image is virtual: the reflected rays are diverging, and the image is where those rays are extrapolated back to. It cannot be formed on a screen.

Same size, 15 cm behind the mirror, virtual.

Section 6: Multiple choice

Paper 1 and Paper 2 style. Choose one answer, A, B, C or D. 8 marks, one for each question.

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1. A ray of light meets a plane mirror. The angle between the ray and the mirror is 62°. What is the angle of incidence?

A 28°
B 56°
C 62°
D 118°

Answer: _____

2. An object is 12 cm in front of a plane mirror. Which statement describes the image?

A It is real, and it lies in the plane of the mirror.
B It is real and the same size, 12 cm behind the mirror.
C It is virtual and the same size, 24 cm behind the mirror.
D It is virtual and the same size, 12 cm behind the mirror.

Answer: _____

3. A ray of light in glass meets a straight glass–air boundary. The angle of incidence is greater than the critical angle. What happens to the ray?

A A refracted ray still leaves the glass and bends away from the normal.
B The ray is totally internally reflected and stays in the glass.
C The ray is totally internally reflected only if it is travelling from air into the glass.
D The ray crosses the boundary in a straight line.

Answer: _____

4. Which statement about red light and violet light is correct?

A Red light has a lower frequency and a longer wavelength than violet light.
B Red light has a higher frequency than violet light.
C Violet light has a longer wavelength than red light.
D Red light and violet light have the same frequency.

Answer: _____

5. A ray of light in air meets a flat glass surface at an angle to the normal. What happens to the speed of the light, and what happens to the ray?

A The speed decreases and the ray bends away from the normal.
B The speed increases and the ray bends towards the normal.
C The speed decreases and the ray bends towards the normal.
D The speed does not change and only the direction changes.

Answer: _____

6. Extended A ray in air meets glass. The angle of incidence is 50° and the angle of refraction is 30°. What is the refractive index of the glass?

A 0.653
B 1.53
C 1.67
D 2.00

Answer: _____

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7. Extended The critical angle for a boundary between a type of glass and air is 34°. What is the refractive index of the glass?

A 0.559
B 1.21
C 1.79
D 2.65

Answer: _____

8. Extended Which object position lets one thin converging lens be used as a magnifying glass?

A The object is further from the lens than 2F.
B The object is at the principal focus.
C The object is between F and 2F, giving an enlarged real image.
D The object is between the lens and the principal focus.

Answer: _____

Section 7: Structured questions

Total: 50 marks. Show your working in every calculation.

1. A ray of light meets a plane mirror. The angle of incidence is 35°.

(a) State the angle of reflection. [1]

(b) Calculate the angle between the incident ray and the mirror. [1]

(c) Extended Describe how to locate the image of a point object in this mirror by a construction that uses two rays. [3]

2. State three characteristics of the image formed by a plane mirror. The object is 8.0 cm in front of the mirror. Include the position of the image. [3]

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3. A student investigates the refraction of light.

(a) Describe an experiment to show refraction by transparent blocks of different shapes. [3]

(b) A ray in air meets a flat face of a rectangular glass block at an angle to the normal, passes through the block, and leaves by the opposite face. Describe what happens to the speed and to the direction of the ray at each face. [3]

4. Light travelling in glass meets a straight glass–air boundary.

(a) State the meaning of the critical angle. [2]

(b) State what happens when the angle of incidence is less than the critical angle, and what happens when it is greater than the critical angle. [2]

(c) Give one everyday use of total internal reflection. [1]

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5. A narrow beam of white light passes through a glass prism.

(a) Describe what is seen, and name the effect. [2]

(b) List the seven colours of the visible spectrum in order of increasing frequency. [1]

(c) State which colour has the longest wavelength, and which colour is refracted the most. [2]

6. Extended The refractive index of a type of glass is 1.6. A ray in this glass meets a straight glass–air boundary at an angle of incidence of 30°.

(a) Calculate the critical angle for this glass. [2]

(b) State whether total internal reflection occurs, and give a reason. [2]

7. Extended Describe how light travels along an optical fibre, and state one use of optical fibres in telecommunications. [3]

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8. A thin lens is placed in a beam of light that is parallel to the principal axis.

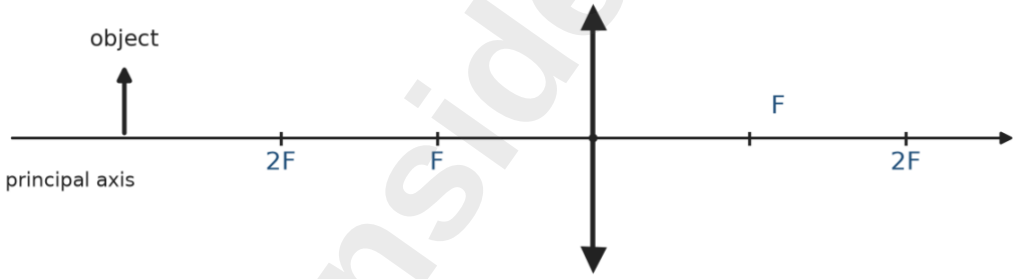
(a) Describe what a thin converging lens does to this beam. [1]

(b) Describe what a thin diverging lens does to this beam. [1]

(c) Define the focal length of a lens. [1]

(d) Define the principal focus of a thin converging lens. [1]

9. The scale diagram shows a thin converging lens, its principal axis, and an object further from the lens than 2F. F and 2F are marked on both sides.

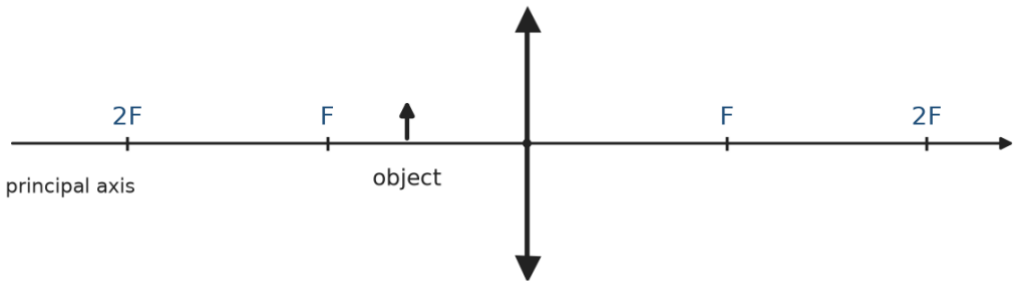


(a) On the diagram, draw two rays from the top of the object and draw the image they locate. [3]

(b) Describe the image. Use the words real or virtual, upright or inverted, and enlarged, same size or diminished. [2]

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10. Extended The scale diagram shows a thin converging lens. The object is between the lens and the principal focus F.



(a) On the diagram, draw two rays from the top of the object to locate the image. Use dotted lines for any rays you extrapolate, and draw the image. [3]

(b) Describe the image, and name this use of the lens. [2]

(c) State the type of lens used to correct long-sightedness, and the type used to correct short-sightedness. [2]

11. Extended A lamp produces visible light of one frequency. The refractive index of a glass block is 1.50, and the speed of light in air is 3.0×10^8 m/s.

(a) State whether this light is monochromatic, and give the meaning of monochromatic. [1]

(b) Calculate the speed of light in the glass. [2]

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Section 8: Examiner tips

Where a tip comes from a Cambridge examiner report for 0625, the exam series and paper are given in brackets.

- Measure every angle of incidence, reflection and refraction from the normal, not from the surface. From air into glass, take the angle between the ray and the surface away from 90° before you use it. The November 2021 Paper 42 report records candidates who did not know that the angle of incidence and the angle of refraction are both measured between the normal and the ray of light. (Nov 2021, Paper 42)
- A plane-mirror image is virtual, the same size as the object, and as far behind the mirror as the object is in front. Many candidates, on the November 2022 Paper 13 question, believed that the image formed by a plane mirror is in the plane of the mirror itself. (Nov 2022, Paper 13)
- Total internal reflection happens only for a ray that is already in the denser medium, such as glass or water, when the angle of incidence at the boundary with air is greater than the critical angle. There is then no refracted ray. On the March 2025 Paper 22 question, the majority believed that the ray had to be travelling from air into water. (March 2025, Paper 22)
- When a question asks for a definition, give the definition, not the condition. The critical angle is the angle of incidence in the denser medium for which the angle of refraction is 90°. Total internal reflection is the reflection of all the light back into the denser medium. "The angle of incidence is greater than the critical angle" is the condition for total internal reflection, not a definition of either. In the November 2022 Paper 43 report, weaker candidates tried to explain the conditions for TIR rather than giving a definition. (Nov 2022, Paper 43)
- Focal length is the distance from the centre of the lens to one principal focus. On the March 2024 Paper 32 question, a common error was to determine the distance between the two focal points. (March 2024, Paper 32)
- A ray through the optical centre of a thin lens is not bent. On the June 2024 Paper 31 question, the most common error was to draw the ray through the centre of the lens deviated along the principal axis. (June 2024, Paper 31)
- Monochromatic light is light of a single frequency. The June 2023 Paper 41 report says candidates could also correctly express this as light of a single wavelength, and that light of one colour can still have a range of frequencies, so "one colour" is an inadequate definition. (June 2023, Paper 41)
- A magnifying glass is one converging lens used with the object between the lens and the principal focus. The image is virtual, upright and enlarged. On the June 2023 Paper 23 question, a common incorrect response showed an enlarged, but real, image. (June 2023, Paper 23)
- Long-sightedness means distant objects are clear and near objects are not, and a converging lens corrects it. Short-sightedness is the reverse, and a diverging lens corrects it. On the June 2024 Paper 41 question, a common misconception was to think that long-sightedness is when people see faraway objects less clearly. (June 2024, Paper 41)

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